

RISHANTH RAJENDHRAN

<https://rishanthrajendhran.github.io/>

rishanth at **umd** dot **edu**

EDUCATION

University of Maryland

College Park, Maryland, US

PhD in Computer Science

Advisor: Mohit Iyyer

Jan 25 - present

University of Massachusetts

Amherst, Massachusetts, US

PhD in Computer Science

Advisor: Mohit Iyyer

Aug 24 - Dec 24

University of Utah

Salt Lake City, Utah, US

Master of Science in Computer Science

Advisor: Ana Marasovic

Aug 22 - May 24

IIT Madras

Chennai, Tamil Nadu, India

Bachelor of Technology in Computer Science and Engineering

Jul 18 - May 22

PUBLICATIONS

1. **Rishanth Rajendhran**, Jenna Russell, Mohit Iyyer, John Wieting. POLARIS: Guiding Small Models to Write Long Stories. Preprint
2. Jenna Russell, **Rishanth Rajendhran**, Chau Minh Pham, Mohit Iyyer, John Wieting. StoryScope: Investigating idiosyncrasies in AI fiction. Preprint
3. **Rishanth Rajendhran**, Amir Zadeh, Matthew Sarte, Chuan Li, Mohit Iyyer. VeriFastScore: Speeding up long-form factuality evaluation. Findings of EMNLP '25
4. Ashim Gupta, **Rishanth Rajendhran**, Nathan Stringham, Vivek Srikumar, Ana Marasović. Whispers of Doubt Amidst Echoes of Triumph in NLP Robustness. NAACL '24

SKILLS

Programming/Scripting Languages Python, C, C++, Javascript, SQL, OWL, HTML, CSS

Frameworks/Tools/Libraries HuggingFace, PyTorch, Pandas, Numpy, Tensorflow, ReactJS, Git, wandb

Natural Languages Tamil, English

RESEARCH EXPERIENCES

Research Assistant

CLIP at UMD

College Park, Maryland, US

Jan 25 - Present

Supervisor: Prof. Mohit Iyyer

VeriFastScore: Evaluating factuality of long-form LLM generations

CoTLM: Teaching LLMs to reason about next token

POLARIS: Training LLMs for long-form creative writing

Research Assistant

UMassNLP

Amherst, Massachusetts, US

Aug 24 - Dec 24

Supervisor: Prof. Mohit Iyyer

Research Assistant
UtahNLP

Salt Lake City, Utah, US
May 23 - Aug 23

Supervisor: Prof. Ana Marasovic

RLHF for NLP: Improved free-text explanations generated by large-scale language models using fine-grained reinforcement learning with human feedback (FG RLHF)
Expressing model uncertainties: Explored introducing model uncertainties in self-rationalization techniques introduced in STaR Self-Taught Reasoner and ZARA Zero-shot Augmentation of Rational-Answer pairs

Student Researcher
UtahNLP

Salt Lake City, Utah, US
Nov 22 - May 23

Supervisor: Prof. Ana Marasovic

Evaluated robustness of large-scale language models on contrastive and out-of-domain datasets
Experimented with evaluation toolkits such as llm-foundry and lm-eval-harness

Undergraduate Researcher
SENAI, IIT Madras

Chennai, Tamil Nadu, India
Aug 21 - Feb 22

Supervisor: Prof. Raghunathan Rengaswamy

Worked on evolving neural networks and constrained models
Investigated autoML and EigenGame

TEACHING EXPERIENCES

Teaching Assistant
University of Utah

Salt Lake City, Utah, US
Jan 24 - May 24

Supervisor: Prof. Ana Marasovic
Course: CS 6340 / 5340 Natural Language Processing

Teaching Assistant
University of Utah

Salt Lake City, Utah, US
Aug 23 - Dec 23

Supervisor: Prof. Anna Fariha
Course: CS 6353 / 5353 Deep Learning

Teaching Assistant
University of Utah

Salt Lake City, Utah, US
Jan 22 - May 23

Supervisor: Prof. Daniel Kopta
Course: CS 5350 Database Management Systems

Teaching Assistant
University of Utah

Salt Lake City, Utah, US
Aug 22 - Dec 23

Supervisor: Prof. Ellen Riloff
Course: CS 6340 / 5340 Natural Language Processing

Instructor
Shaastra Tech Fest, IIT Madras

Chennai, Tamil Nadu, India
Jan 22

Conducted two-day workshop on python programming and web and app development

Volunteered to teach programming in C++ to 12th grade students from underprivileged backgrounds
Held theory followed by lab sessions to solidify understanding

OTHER EXPERIENCES

Web Developer Intern

EduVisory, Nirmaan-IITM incubated startup

Remote
Jun 20 - Jul 20; Aug 20 - Jan 21

Supervisor: Dr. Ravi Dadsena

Worked on building website for this ed-tech startup from scratch
Designed user interface of website and assisted the development of backend

App Developer

Product Design Club, Centre for Innovation, IIT Madras

Remote
Aug 20 - Apr 21

Worked on building website and app for streamlining garbage disposal inside the IIT Madras campus

Coordinator

Institute WebOps, IIT Madras

Remote
Jun 20 - Apr 21

Worked on maintaining institute websites
Organized learning tasks for onboarding students

FEATURED PROJECTS

C-STaR

University of Massachusetts Amherst

Github: <https://github.com/vule20/C-STaR>

Fall 24

Incorporating uncertainty in STaR pipeline to improve model-generated reasoning

Salience for Automatic Speech Recognition

University of Utah

Github: <https://github.com/jacobkj314/salASR>

Fall 23

Extended input attribution techniques found in deep learning literature to the task of ASR
Verified the soundness of salience as an explainability method for Whisper models using remove-and-retrain (RoAR) approach

Relation Extraction using Distant Supervision

University of Utah

Github: <https://github.com/RishanthRajendhran/relationExtraction>

Spring 23

Constructed a dataset from Wikipedia using relations in FB15K, a knowledge base completion dataset
Built a BERT-based model for 10-class classification
Built a Tree LSTM-based model for 10-class classification

Robustness Evaluation

University of Utah

Github: <https://github.com/RishanthRajendhran/robustnessEvaluation>

Fall 22 - Fall 23

Evaluated SotA LLMs on contrast sets
Explored greedy-decoding and self-consistency with and without chain-of-thoughts

Visual Question Answering

University of Utah

Github: <https://github.com/RishanthRajendhran/visualQA>

Spring 23

Built an encoder-decoder model with RESNET as the base

Fall 21 - Spring 22

Fall 21 - Spring 22

Replicated results from the paper that introduced EigenGame
Conducted experiments and investigated effects of various initializations and gradient ascent approaches using 3D visualizations and plots

Transliteration using RNNs

Spring 21

IIT Madras

Github: <https://github.com/RishanthRajendhran/transliterationUsingRNNs>

Built a Tamil-to-English transliteration system
Investigated attention mechanism using gradient-based attribution by visualizing attention heatmaps
Identified weaknesses of the model by inspection of plots

CERTIFICATIONS

Amazon ML Summer School

Online
July 2021

Interacted with Amazon ML Engineers; discussed practical issues with using ML models on real world applications

NVIDIA Machine Learning and Deep Learning

Online
June 2021

Workshop on choosing the right ML model for given data, transformer architecture. based
Natural Language Processing applications
Course covered machine learning models and techniques